TEXORA · LECTURE SHEET

Environmental Challenges

What the environment is, how its parts hold together, and the major challenges — soil erosion, desertification, biodiversity loss, deforestation, climate change, effluents and their health effects — that frame everything else in this course.

TE-4204 · ENVIRONMENTAL ENGINEERING AND POLLUTION CONTROL · TOPIC 1 · LEVEL 4 TERM 2

1. What is the environment?

The **environment** is the totality of the external conditions — physical, chemical, biological and social — that surround a living organism or a community and influence its survival, growth and behaviour. For an environmental engineer the useful working definition is: *the environment is the interacting system of air, water, land, living organisms and human-made surroundings within which all industrial and natural processes take place.* Nothing a factory discharges disappears; it only moves into one of these parts of the system, and from there it can move into the others.

Four great spheres make up the physical environment. The **atmosphere** is the gaseous envelope (air) that supplies oxygen and carbon dioxide and regulates temperature. The **hydrosphere** is all water — oceans, rivers, groundwater, rain. The **lithosphere** is the solid earth: soil and rock that supply minerals and receive solid waste. The **biosphere** is the zone of life, where the other three spheres overlap and organisms live. The spheres exchange matter constantly: rain washes air pollutants into rivers, river silt rebuilds land, soil nutrients feed plants, and plants return oxygen to the air. This is why a pollution problem created in one sphere — dyehouse effluent in water, boiler smoke in air — rarely stays in that sphere.

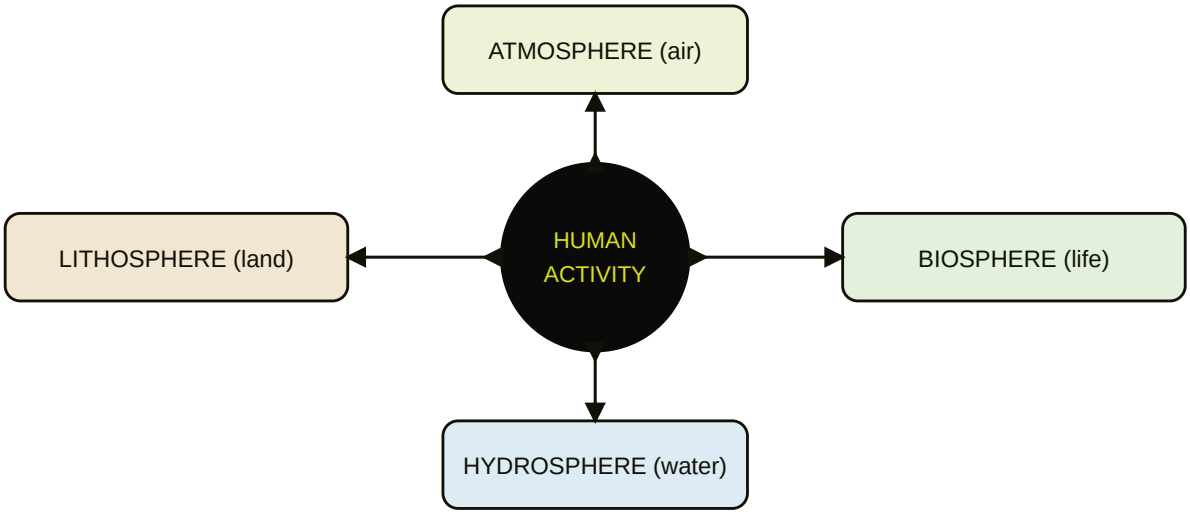


Figure 1 — The four spheres of the environment and human activity at the centre. The double-headed arrows are the key idea: matter and energy move in both directions, so pollution released to any one sphere spreads to the rest.

2. Soil erosion and desertification

Soil erosion is the removal of topsoil by water, wind, tillage and the loss of protective vegetation. Topsoil holds most of the nutrients and organic matter a crop needs, and it forms extremely slowly — a centimetre of topsoil may take centuries to build — so erosion is effectively a permanent loss on a human timescale. Water erosion dominates in monsoon climates such as Bangladesh, where intense rain on bare or freshly tilled land washes soil into rivers; the silt then raises river beds and worsens flooding downstream. Wind erosion dominates in dry regions and on loose, sandy soils.

Desertification is the degradation of land in dry areas until it loses its productive capacity: vegetation cover thins, soil organic matter falls, the surface crusts or turns to moving sand, and wells dry up. Its causes combine climate stress with human pressure — over-cultivation, over-grazing, deforestation and poor irrigation that leaves salts behind. For a textile engineer the connection is raw material: cotton, jute and other natural fibres are agricultural products, and land degradation in fibre-growing regions shows up as falling yields and rising raw-material prices. Desertification is fought with the same tools everywhere — vegetation cover, windbreaks, controlled grazing, water harvesting and soil organic-matter restoration.

3. The pest-control problem

Pests destroy a significant share of the world's crops, and cotton is one of the most pest-attacked crops of all. Chemical **pest control** with synthetic pesticides solved the immediate problem but created three environmental ones. First, **resistance**: repeated spraying selects insects that survive it, forcing higher doses or new chemicals — the "pesticide treadmill". Second, **non-target effects**: broad-spectrum

insecticides kill the predators and pollinators that naturally restrain pests, sometimes causing worse outbreaks ("resurgence") of the pest or of secondary pests. Third, **residues**: persistent compounds such as the older organochlorines accumulate through food chains and have been found in soil, water and human tissue long after use stopped. The modern answer is **integrated pest management (IPM)** — monitoring pest levels, using biological controls (parasites, predators, pheromone traps), resistant crop varieties and targeted, lower-toxicity chemicals only when thresholds are crossed. The textile link is direct: pesticide residues on raw cotton are one reason buyers now demand certified fibre and restricted-substance testing of finished goods.

4. Biodiversity loss and deforestation

Biodiversity — the variety of genes, species and ecosystems — is the living library from which agriculture, medicine and industry draw. It is being lost mainly through habitat destruction, over-exploitation, pollution and climate change. Each extinction is irreversible, and the loss of pollinators, soil organisms and wild crop relatives weakens the farming systems the textile industry depends on.

Deforestation — the permanent conversion of forest to other land uses — drives much of this loss and also drives climate change: forests store carbon in wood and soil, and clearing releases it. Deforestation increases soil erosion and flood peaks, reduces dry-season river flow, and removes the watershed protection that keeps reservoirs and irrigation supplies usable. Causes include agricultural expansion, fuelwood collection, logging and infrastructure. Countermeasures are protection of remaining forest, reforestation and agroforestry, alternative fuels, and certification schemes that let buyers choose timber, paper and cellulosic fibres (such as viscose) from responsibly managed forests.

5. Water pollution, fisheries and large dams

Water pollution receives a full topic later in this course (Topic 6); here it belongs in the challenge list. Rivers and aquifers receive industrial effluent, agricultural run-off carrying fertiliser and pesticide, and untreated sewage. The consequences fall first on aquatic life: oxygen-demanding waste suffocates fish, toxic substances accumulate in their tissues, and nutrient enrichment (eutrophication) chokes water bodies with algae.

Fisheries resources are therefore an environmental indicator as well as a food source. Bangladesh's rivers, floodplains (haors and beels) and the Bay of Bengal support one of the world's largest inland fisheries; declining catches and disappearing species are widely read as symptoms of polluted and obstructed waterways — blocked migration routes, industrial discharge into rivers such as the Shitalakshya and Buriganga, and loss of floodplain breeding habitat.

Large dam projects show why every environmental intervention has two sides. Dams store water for irrigation, generate hydroelectricity and moderate floods; but they also trap silt that would fertilise

downstream farmland, block fish migration, submerge land and settlements, change downstream flow patterns that deltas depend on, and can increase water-borne disease around reservoirs. Modern practice therefore requires an **environmental impact assessment (EIA)** before approval, weighing these effects openly rather than discovering them afterwards — a principle that returns in Topic 8.

6. Climate change and urbanization

Climate change — the long-term warming of the planet caused mainly by greenhouse-gas accumulation (Topic 2) — expresses itself as rising average temperatures, shifting rainfall, more intense extreme events, and sea-level rise. For a low-lying delta country the front-line effects are salinity intrusion into farmland and drinking water, stronger cyclones, river-bank erosion and pressure on the very regions that grow cotton and jute and house much of the garment industry. Industry is both a contributor (energy use, process emissions) and a victim (flooded factories, disrupted supply chains), which is why buyers increasingly ask suppliers for energy and emissions data.

Urbanization concentrates people, industry and waste. Planned cities gain from shared water, sewerage and treatment infrastructure; unplanned industrial clustering — the pattern around many textile zones — overwhelms it, so effluent, solid waste, air emissions and noise all reach residents at once. The environmental challenge of the city is therefore largely a challenge of infrastructure and enforcement keeping pace with growth.

7. Effluents and related health issues

An **effluent** is any liquid waste stream discharged from a process. Textile effluents are among the most complex industrial wastewaters: high in organic load (BOD/COD), colour, dissolved salts, and processing chemicals, and variable from hour to hour as production changes. Health effects follow exposure routes: contaminated drinking water spreads enteric disease and, with toxic chemicals, chronic poisoning; air emissions irritate lungs and eyes; noise damages hearing (Topic 5); unsafe solid waste exposes workers and waste-pickers. The chain is always the same — **source → pathway → receptor** — and environmental engineering works by breaking the chain at its cheapest, most reliable point, usually the source.

CHALLENGE	MAIN DRIVERS	KEY CONSEQUENCES	TEXTILE-INDUSTRY LINK
Soil erosion	Deforestation, over-tillage, intense rain	Fertility loss, river siltation, floods	Fibre-crop yields fall
Desertification	Over-use of dry lands, climate stress	Land loses productivity	Cotton-growing regions at risk

Pest control	Broad-spectrum pesticide dependence	Resistance, residues, predator loss	Residues on raw cotton; buyer testing
Biodiversity loss	Habitat loss, pollution, over-use	Species extinction, weaker ecosystems	Loss of fibre plants and pollinators
Deforestation	Clearing for farms, fuel, timber	Carbon release, erosion, flow changes	Cellulosic fibre sourcing; paper/packaging
Water pollution	Effluent, sewage, agrochemical run-off	Fish kills, unsafe water, eutrophication	Wet-processing discharge
Climate change	Greenhouse-gas accumulation	Heat, extremes, sea-level rise, salinity	Energy cost, supply-chain disruption
Urbanization	Unplanned industrial growth	Waste and emissions overwhelm services	Industrial clusters near cities

Glossary

Environment — the total external conditions surrounding an organism or community. **Atmosphere / hydrosphere / lithosphere / biosphere** — the air, water, land and life spheres. **Soil erosion** — removal of topsoil by water, wind or tillage. **Desertification** — degradation of dry land to a desert-like, unproductive state. **Biodiversity** — variety of genes, species and ecosystems. **Deforestation** — permanent conversion of forest to non-forest use. **Effluent** — liquid waste discharged from a process. **Eutrophication** — nutrient over-enrichment of water causing excessive algal growth and oxygen loss. **Integrated pest management (IPM)** — combined biological, cultural and targeted chemical pest control. **Environmental impact assessment (EIA)** — systematic study of a project's environmental effects before approval.

Key points

• The environment is one interacting system of four spheres; pollution released to any sphere spreads to the others.

• Soil erosion and desertification destroy productive land slowly and almost irreversibly — and fibre crops grow on that land.

• Chemical pest control creates resistance, non-target damage and residues; IPM is the modern corrective.

• Biodiversity loss and deforestation weaken ecosystems and release stored carbon.

• Water pollution shows up early as damage to fisheries; large dams trade real benefits against real ecological costs and require EIA.

• Climate change and unplanned urbanization multiply every other challenge; effluent health effects follow the source → pathway → receptor chain.

Practice

1. Define environment in your own words and name its four physical spheres, giving one textile example of matter moving between two of them.
2. Explain why topsoil erosion is treated as almost irreversible on a human timescale, and list three control measures.
3. What is the "pesticide treadmill"? Describe two further environmental problems of broad-spectrum pesticide use on cotton.
4. Give two benefits and three environmental costs of a large dam, and state the assessment tool used before such a project is approved.
5. Trace the source → pathway → receptor chain for dyehouse effluent reaching a fishing community.

Answer key

1. Environment = the totality of external physical, chemical, biological and social conditions affecting an organism. Spheres: atmosphere, hydrosphere, lithosphere, biosphere. Example: boiler ash (from air emissions) is washed by rain into a river — atmosphere to hydrosphere; or river silt deposited on farmland — hydrosphere to lithosphere.
2. Topsoil forms over centuries but can be stripped in a few seasons of heavy rain on bare land, so the loss cannot be made good within a farming lifetime. Controls: maintain vegetation cover/contour planting or terracing; reduced tillage and mulching; drainage and check dams to slow run-off (any three).
3. The treadmill: spraying kills most insects but the resistant survivors breed a resistant population, so doses or toxicity must keep rising to hold the same control. Other problems: killing of natural predators and pollinators causing pest resurgence and secondary-pest outbreaks; persistent residues accumulating in soil, water, food chains and fibre.
4. Benefits: irrigation storage/flood moderation and hydroelectric generation. Costs: silt trapped upstream (downstream fertility and delta maintenance suffer), fish migration blocked, land and settlements submerged (also acceptable: changed downstream flow, disease around reservoirs). Tool: environmental impact assessment (EIA).
5. Source: untreated or poorly treated dyeing effluent discharged to a canal/river. Pathway: river water carries colour, organic load and salts downstream; fish absorb/accumulate contaminants and dissolved oxygen falls. Receptor: the fishing community — reduced and contaminated catch, unsafe water for domestic use, income and health losses.